

Kariuki Kelvin Wachira

Full-Stack Developer · Mobile Security · Cloud Infrastructure

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PROFESSIONAL SUMMARY

BSc. Undergraduate Information Technology student at the University of Embu and Power Learn Project (PLP) graduate student. A problem solver with hands-on experience in designing and shipping production-grade full-stack web and mobile applications, and being an independent problem solver, managing every layer, including database schema and server infrastructure, to UI and cloud deployment. Interested in Mobile Security, Hybrid End-to-End Encryption, real-time systems and constitution-driven software design. Member of the Google Developer Group (GDG) at the University of Embu. Offers freelance web development and technical support to local clients and has a 5-star Google rating due to the consistent delivery and technical quality.

EXPERIENCE

IT Attaché Intern — Chinga Tea Factory Ltd

2nd May 2025 – 2nd August 2025

- Operated the ERP system (SAP) supporting core business and internal data management workflows
- Used POS scanners and field data collection tools to process tea leaf deliveries from farmers
- Processed farmer loan applications and disbursements via the GreenFedha platform
- Issued insurance covers to farmers and their successors through the Majani Insurance system
- Manned the customer care desk - responding to farmer questions and solving account problems.

EDUCATION

BSc. Information Technology · University of Embu 2022-2026

Currently Enrolled, awaiting graduation

Focus areas: Network Security, Mobile Development, Data Structures & Algorithms

Full-Stack Web Development Certificate · Power Learn Project (PLP) 2025

Intensive program covering Python, JavaScript, SQL, and modern web frameworks

Kenya Certificate of Secondary Education (KCSE) 2018- 2021

Grade C+ 20

Kenya Certificate of Primary Education (KCPE) 2010- 2017

Muirungi Primary School

TECHNICAL SKILLS

- Python
- JavaScript
- MySQL / SQL
- Data Visualization
- Node.js / Express
- Django
- Java
- Flutter / Dart
- Cybersecurity
- React / Next.js

Cloud & DevOps: AWS (EC2, S3) · Nginx · PM2 · Netlify · Supabase · PostgreSQL

Tools & Protocols: Socket.io (WebSockets) · Prisma ORM · JWT / bcrypt · REST APIs · Git/GitHub · Android Studio

Security: RSA/AES Hybrid Encryption · End-to-End Encryption (E2EE) · Mobile Security · JWT Auth

FEATURED PROJECTS

C2C Marketplace Platform — *Full-Stack Real-Time Web Application*

A production-ready Consumer-to-Consumer marketplace, built from scratch with a fully decoupled architecture. Allows individuals to list items, browse a live feed, and negotiate via real-time private chat all without third-party apps.

- Architected a decoupled full-stack system: Next.js 15 (App Router) frontend on Netlify CDN + Node.js/Express backend on AWS EC2 behind Nginx reverse proxy, enabling independent scaling and CI/CD auto-deploy
- Implemented real-time bidirectional communication using Socket.io (WebSockets) with live typing indicators and JWT-authenticated WebSocket connections — messages delivered in milliseconds without polling
- Designed a PostgreSQL schema (Users, Products, OrderInterests, Messages) managed via Prisma ORM with type-safe client generation; hosted on Supabase with PgBouncer connection pooling
- Secured auth with bcrypt (cost factor 12) + stateless JWT, enabling horizontal scaling; live camera capture for product photos uploaded directly to AWS S3 (UUID filenames, 5 images / 10 MB each)
- Deployed with PM2 for zero-downtime process management and automatic crash recovery; 12+ REST endpoints, 5 Socket.io event types, 7 routes, 400 ms search debounce, 12-item paginated feed

[Next.js 15](#) · [Node.js](#) · [Express.js](#) · [Socket.io](#) · [PostgreSQL](#) · [Prisma ORM](#) · [AWS EC2](#) · [AWS S3](#) · [Supabase](#) · [Nginx](#) · [PM2](#) · [JWT](#) · [bcrypt](#) · [Tailwind CSS](#) · [Zustand](#)

NivlockX — *Secure Mobile File Storage & Sharing App (Android) using Hybrid E2EE encryption*

An Android application implementing Hybrid End-to-End Encryption (E2EE) for secure file storage and peer-to-peer sharing. Combines AES symmetric encryption with RSA asymmetric key exchange — the same cryptographic architecture used by Signal and WhatsApp.

- Implemented RSA-2048 for key exchange and AES-256 for bulk file encryption — no plaintext data is stored or transmitted at any point in the pipeline
- Built natively in Flutter/Dart targeting Android; encryption and decryption are performed entirely on-device before any network transmission, ensuring zero server-side exposure
- APK available for direct download, demonstrating end-to-end delivery from development to distribution

[Flutter](#) · [Dart](#) · [AES-256](#) · [RSA-2048](#) · [E2EE](#) · [Android Studio](#) · [Mobile Security](#)

AQUAMY — Chama Management System — *Full-Stack Financial & Administrative Platform*

A full-stack web application that digitises and automates the complete financial and administrative operations of a registered Kenyan community savings group (Chama). Replaces paper ledgers, WhatsApp coordination, and spreadsheets with a constitution-governed digital platform encoding fee amounts, age limits, loan policies, fine schedules, and role permissions directly into the application logic.

- Designed a 17-model PostgreSQL schema mapping directly to the group's constitution: members, contributions, full loan lifecycle (DRAFT → DISBURSED → FULLY_REPAID), versioned loan policies, guarantors, meetings, attendance, fines, shares, and an append-only group treasury ledger
- Implemented a 9-role RBAC system (Admin, Chairperson, Secretary, Treasurer, Credit Committee, etc.) with all permissions enforced server-side via NextAuth.js JWT sessions and Next.js Server Actions
- Built full M-Pesa Daraja API integration (STK Push + callback handling) for contributions, loan repayments, fines, and share purchases — raw callback payloads preserved for dispute resolution
- Engineered automatic fine creation from meeting attendance (LATE → KES 50, ABSENT → KES 20), quorum enforcement, and a fine waiver audit trail (waivedBy, waivedAt, reason) for compliance
- Implemented a MemberFinancialSummary denormalised aggregate model for O(1) dashboard reads, and an append-only GroupLedgerEntry table for real-time treasury balance without expensive live SUM() queries
- Built an immutable AuditLog for every sensitive operation (loan approvals, role changes, suspensions) with old/new JSON snapshots; InviteCode system prevents unauthorised member registration

[Next.js 14](#) · [TypeScript](#) · [NextAuth.js](#) · [Prisma ORM](#) · [PostgreSQL](#) · [Supabase](#) · [M-Pesa Daraja API](#) · [Tailwind CSS](#) · [AWS EC2](#) · [Docker](#) · [Server Actions](#)

COMMUNITY & EXTRACURRICULAR

Google Developer Group (GDG) — University of Embu Chapter

Active member of the Google Developer Group on campus, participating in tech talks, workshops, and collaborative sessions focused on modern software development, cloud technologies, and developer tooling.

Nivlek Solutions — Web Development & IT Services

Independently provide web development, SEO, and technical support services to local clients under the Nivlek Solutions brand. Rated 5 stars on Google Reviews.

KEY COMPETENCIES

System Architecture & API Design · Real-Time Systems (WebSockets) · Cloud Deployment (AWS, Netlify) · End-to-End Encryption · Mobile Development (Flutter/Android) · Database Design (PostgreSQL/MySQL) · Secure Authentication (JWT/bcrypt) · Data Visualization · IT Consultancy

REFERENCES

Dr. Victoria Mukami

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Mr. Peter Macharia

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